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IMMUNITY NUTRITION POPULAR DIETS

CLINICAL NUTRITION TRAINING WEBINAR

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SYNOPSIS



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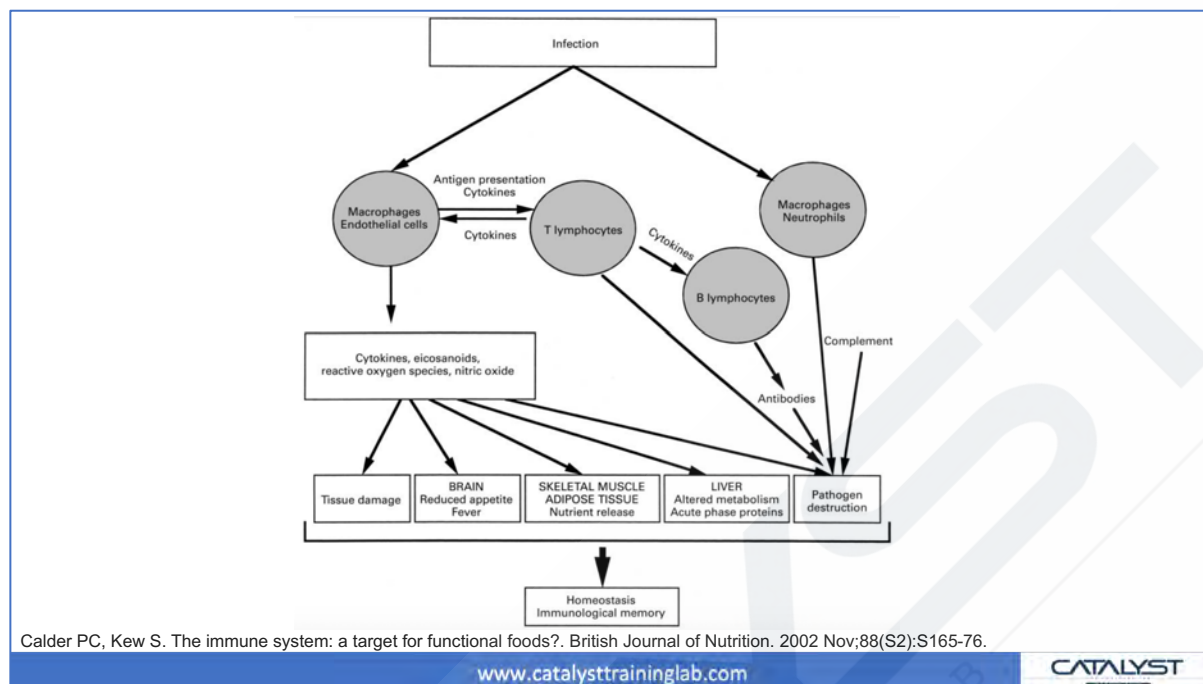
Table of Contents

Overview	2
Nutrition & Immunity.....	3
Malnutrition, Under-nutrition & Immunity.....	9
Dietary patterns to support immune system	11
Mediterranean Diet	11
Vegetarian/ Plant-based Diet	12
Keto/Low Carb Diet	12
Very-Low Calorie Diet (VLCD)	13
Other factors that boost the immune system	13

Overview

The immune system is a complex ensemble of biological entities (cells, tissues, organs) and processes (inflammation, immune tolerance) that protect the integrity of the organism from external and internal threats.

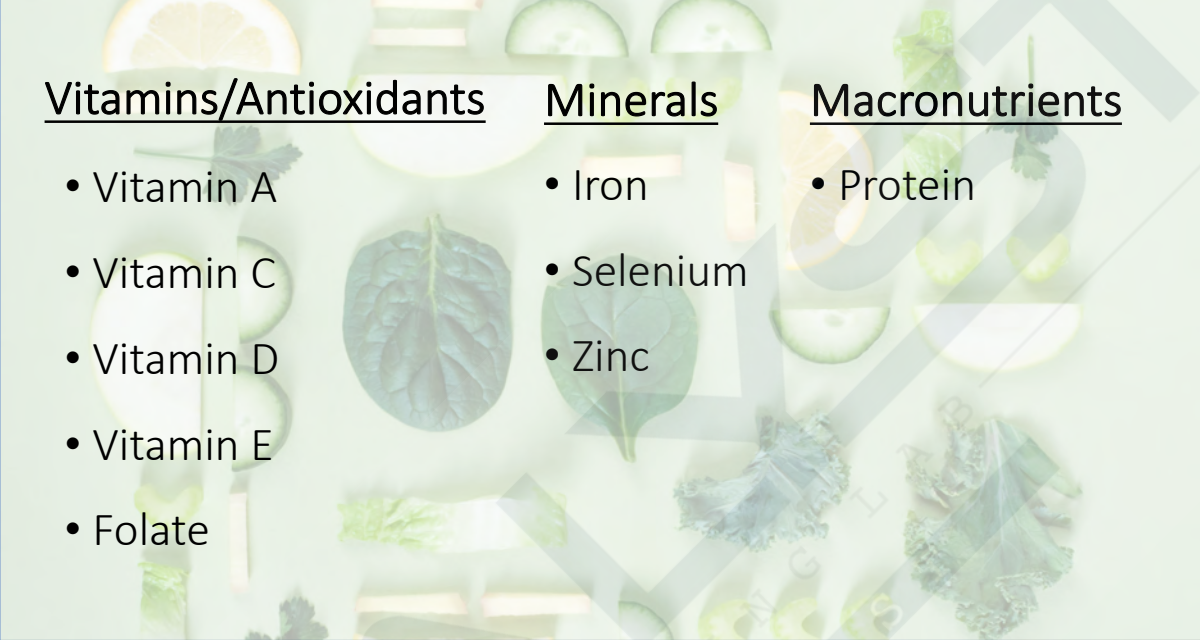
External threats include micro-organisms (bacteria, viruses, parasites), their toxic products (exotoxins, endo-toxins, enzymes), and air- and food-borne allergens. The immune system also responds to severe trauma such as burns and physical injuries in a manner similar to the shock response that occurs with an overwhelming bacterial infection. There are two major functional components of the immune system, the innate and the adaptive. Innate immunity is present prior to exposure to antigens and accordingly not customized. The innate is designed to recognize a few highly conserved motifs present in microorganisms. In contrast, the adaptive immune system is highly specific and increases both in specificity and magnitude with repeated exposure to antigen. Unlike the innate that recognizes pathogens through specific molecular markers, the cells of the adaptive immune system need to be educated to discriminate between self and non-self.



Nutrition & Immunity

The nutritional status plays an important role in the functioning of the immune system. Proteins, carbohydrates and fats, as well as micronutrients interact with immune cells systematically in blood, regional lymph nodes and in the specialized gastrointestinal immune system. Carbs can be considered primarily the source of energy in immune-nutrition, while amino acids such as glutamine, arginine, taurine, and sulfur-containing amino acids have immune-modulatory effects. Declines in both specific and non-specific immunity have been reported in association with under-nutrition and protein deficiency. There is also considerable evidence that deficiencies of trace elements such as iron, zinc, selenium and copper, and vitamins A, B6, B12, folic acid, C, D and E are associated with impairments in immune function. While natural food has the potential to supply most of the essential macro- and micronutrients, dietary supplements and/or enriched foods might be of great value in

stress situations such as premature life, ageing or disease, or extreme conditions like exercise.



<u>Vitamins/Antioxidants</u>	<u>Minerals</u>	<u>Macronutrients</u>
• Vitamin A • Vitamin C • Vitamin D • Vitamin E • Folate	• Iron • Selenium • Zinc	• Protein

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Recommended Daily Allowance (RDA) Vitamin A

Age	Male (µg=mcg/day)	Female (µg/day)
0-12 mo	400-500	400-500
1-13 yr	300-600	300-600
14-18 yr	900	700
19-70 yr	900	700
Pregnancy/lactation 18+ yr	—	750 for pregnancy 1300 for lactation

<https://lpi.oregonstate.edu/mic/vitamins/vitamin-A>

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Vitamin D RDA

Age	Male (IU/day)	Female (IU/day)
0-12 mo	400	400
1-13 yr	600	600
14-18 yr	600	600
19-70 yr	600	600 (also for pregnancy/lactation)
70+ yr	800	800

<https://ods.od.nih.gov/factsheets/VitaminD-HealthProfessional/>

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Iron RDA

Age	Male (RDA)	Female	Pregnancy	Lactation
0-6 mo	0.27mg	0.27g	NA	NA
7-12 mo	11 mg	11mg	NA	NA
1-3 yr	7mg	7mg	NA	NA
4-8 yr	10mg	10mg	NA	NA
9-13 yr	8mg	8mg	NA	NA
14-18 yr	11mg	15mg	27mg	10mg
19-50 yr	8mg	18mg	27mg	9mg
51+ yr	8mg	8mg	NA	NA

<https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/>

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Vitamin E RDA

Age	Male (mg/day)	Female (mg/day)
0-12 mo	5	5
1-13 yr	6-11	6-11
14-18 yr	15	15
19-70 yr	15	15
Pregnancy/lactation All ages	—	15 for pregnancy 19 for lactation

<https://lpi.oregonstate.edu/mic/vitamins/vitamin-E>

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Vitamin C RDA

Age	Male (mg/day)	Female (mg/day)
0-12 mo	40-50	40-50
1-13 yr	15-45	15-45
14-18 yr	75	65
19-70 yr	90	75
19+ yr Smokers	125	110
Pregnancy/lactation 18+ yr	—	85 for pregnancy 120 for lactation

<https://lpi.oregonstate.edu/mic/vitamins/vitamin-C>

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Folate RDA

Age	Male (µg/day)	Female (µg/day)
0-12 mo	65-80	65-80
1-13 yr	150-300	150-300
14-18 yr	400	400
19-70 yr	400	400
Pregnancy/lactation All ages	—	600 for pregnancy 500 for lactation

<https://lpi.oregonstate.edu/mic/vitamins/folate>

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Selenium RDA

Age	Male (µg/day)	Female (µg/day)
0-12 mo	15-20	15-20
1-13 yr	20-40	20-40
14-18 yr	55	55
19-70 yr	55	55
Pregnancy/lactation All ages	—	60 for pregnancy 70 for lactation

<https://lpi.oregonstate.edu/mic/minerals/selenium>

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Zinc RDA

Age	Male	Female	Pregnancy	Lactation
0-6 mo	2 mg	2 mg	NA	NA
7-12 mo	3 mg	3 mg	NA	NA
1-3 yr	3 mg	3 mg	NA	NA
4-8 yr	5 mg	5 mg	NA	NA
9-13 yr	8 mg	8 mg	NA	NA
14-18 yr	11 mg	9 mg	12 mg	13 mg
19+ yr	11 mg	8 mg	11 mg	12mg

<https://ods.od.nih.gov/factsheets/Zinc-HealthProfessional/>

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Sample Menu Providing Key *Vitamins*

Meal	Foods	Vitamins
Breakfast	Fortified breakfast cereal with Strawberries Pumpkin or sunflower seeds 2% milk (or fortified non-dairy milk)	Vitamin A, D, folate Vitamin C, folate Vitamin E Vitamin D
Lunch	Vegetable soup with whole grain pasta Salad with chickpeas and quinoa	Folate, vitamin A, C, E Folate, vitamin A, C, E
Dinner	Tofu stir fry with vegetables + Baba Ghanoush + Fattoush Salad	Vitamin A, C, E Folate, Vitamin C, E
Snack	Orange Almonds Greek yogurt	Vitamin C Vitamin E Vitamin D

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Malnutrition, Under-nutrition & Immunity

Malnutrition and infection are two major obstacles for health, development and survival, with malnutrition being the commonest cause of immunodeficiency at a global scale. Nutritional deficiency is associated with impaired immune responses, especially cell-mediated immunity, cytokine production, secretory antibody response and affinity. The proper consumption and absorption of micronutrients is essential for optimal immune response (zinc, iron, selenium, vitamin A, pyridoxine, vitamin E). But macronutrient balance also plays a role. There are subtle differences in immune responses of animals fed casein- and whey-based diets for example. During periods of stress and illness, production of glutamine—the most abundant intracellular amino acid—is upregulated as branched chain amino acids are metabolized by skeletal muscle. Glutamine is an important energy source for intestinal enterocytes and for rapidly proliferating cells such as immunocytes which react to challenges imposed by injury and illness. The immune system is undergoing permanent renewal and produces millions of immune cells daily. Immune cell renewal is elevated during infectious disease, and recovery depends on the rate of cell division between the invading microorganism and that of immune cells. The immune system uses both macro-and micronutrients involved in DNA, RNA and protein synthesis. Thus, under-nutrition has a strong influence on the immune system at all ages but mainly in growing and aged humans, i.e. when the body's nutritional reserves are limited. At those life stages, under-nutrition is a major factor leading to immunodeficiency and thereby to higher infection rates.

Protein

Protein		Additional Properties of Protein Related to Immune Function
Function	Promotes cell reproduction and tissue growth/repair and wound healing. Essential for making DNA. Supports the immune system/making antibodies.	- The foundation of immuno-protection via immunoproteins: - Immunoglobulins (Ig) and antibodies (Ab) -IgA, IgM, IgE, IgD - Igs are produced by plasma cells derived from B-lymphocytes - Igs bind to antigens (virus and bacteria) and inactivate them - Cytokines are produced by white blood cells (like T-helper/CD4 cells) and macrophages - Macrophages and neutrophils destroy foreign antigens through phagocytosis
Deficiency /toxicity	Inadequate and excessive protein can cause complications in metabolically stressed individuals.	
Needs	Adults 0.8-1g/kg Hospitalized adults 1-1.2g/kg Hospitalized/stressed: 1.5-2g/kg	
Testing	Not common; can test plasma albumin levels but it's not a reliable indicator of status	

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Protein Food Sources



Food	Serving	Protein
Tofu	½ cup	10g
Soy milk (other plant-based milks?)	1 cup	8g
Tempeh	½ cup	16g
Lentils (all types)	½ cup	9g
Beans (black, pinto, kidney, chickpea, white)	½ cup	7-8g
Pumpkin, flax, sunflower, chia seeds	1 ounce	5-9g
Whole grains (wheat, quinoa, oatmeal)	½ cup	3-6g
Nuts (peanut, almonds, pistachios, hazelnuts, walnuts, pecans, cashews)	1 ounce	4-7g
*Poultry/beef/pork	3 oz	27g
*Egg	1	6g
*Seafood (fish/shellfish)	3oz	22g

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Clinical Takeaways for the Immune System and the Mediterranean Diet

- Include protein in each meal and snack
 - Ensure you're meeting your iron and zinc needs
 - Seafood (as able)
 - Poultry and eggs
- Optimizing whole grain, healthy fats, legumes, fruit and vegetable intake to meet adequate calories and support immune health
 - Whole wheat, quinoa, farro, oats, bulgur etc
 - Utilize beans + lentils
 - Dairy foods (milk, cheese, yogurt) *low fat options if needed
 - Nuts, seeds, avocado, olives, olive oils
 - All fruits and vegetables – eat the rainbow

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Clinical Takeaways for the the Immune System and Vegetarian/Plant-Based Diets

- Assess what foods your vegetarian/vegan clients includes
- Include protein in each meal and snack
 - Ensure clients meet iron and zinc needs
- Optimize whole grain, healthy fats, legumes, fruit and vegetable intake to meet adequate calories and support immune health:
 - Whole wheat, quinoa, farro, oats, bulgur etc
 - Beans, lentils, soy foods (soy milk, tofu, tempeh, edamame, miso)
 - Nuts, seeds, avocado, oils
 - All fruits and vegetables

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Clinical Takeaways for the Immune System and the Ketogenic/Low Carb Diet

- Eat as many carbohydrate rich foods as you're able per the diet
 - If you're willing to go up to 100g/d CHO = more options
 - Typical keto diet = 20-30g/d
- Include protein with every meal and snack
 - Eggs, seafood, poultry, lean red meat, some nuts/seeds, tofu, low carb dairy products. Consider protein powder?
- Maximize fruits and vegetables
 - Leafy greens, broccoli, cauliflower, bell peppers, olives, carrots
 - Avocado, tomatoes, berries, citrus, watermelon, pomegranate

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Clinical Takeaways for the VLCD

- Be sure to meet calorie needs to avoid malnutrition
- This should be medically supervised and structured
- Consider a MVI to cast a safety net of nutrients
- Protein with each meal and snack
- Include fruits and vegetables with every meal and snack if possible
- Assess meal replacement foods and shakes
 - Ensure they are providing adequate nutrients and calories
 - Avoid pro-inflammatory ingredients: corn syrup, artificial sweeteners, omega 6 oils

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Other factors that boost the immune system

Since many of the functions of the immune system are dependent or associated with circadian rhythm, sleep is considered a corner stone for boosting immunity. It is worth mentioning that checking tech-related habits (night chatting, social media surfing) that span late into the night might be considered a key indicator for immune-compromised patient.

Clinical Recommendations for Sleep

Age	Hours of Sleep
Newborns (0-3 mos)	14-17
Infants (4-11 mos)	12-15
Toddlers (1-2 yrs)	11-14
Preschoolers (3-5 yrs)	10-13
School age children (6-13 yrs)	9-11
Teenagers (14-17 yrs)	8-10
Younger adults (18-25 yrs)	7-9
Adults (26-64 yrs)	7-9
Older adults (65+ yrs)	7-8

Sleep Hygiene

- Stick to a sleep schedule, even on weekends
- Practice a relaxing bedtime routine
- Exercise daily
- Evaluate bedroom for ideal temperature, sound, and light
- Sleep on comfortable mattress and pillows
- Be aware of alcohol and caffeine intake
- Avoid electronic use before bed

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The other factor includes stress, which strongly impacts immunity in relation to duration of stressors. Stress can be categorized into acute and chronic with negative repercussions for each on the patients.

Clinical Recommendations for Stress Management

To reduce stress, clients should:

- Be aware of initial signs of stress reaction
- Develop basic stress management skills and apply them to real life

Stress management skills:

- Relaxation: breathing techniques, biofeedback, etc
- Cognitive techniques: restructure your thinking, schedule, positive imagery, set goals
- Behavioral changes: check your assumptions, share your expectations, exercise and drink sensibly
- Relationship review: review past hurts, forgive, communicate feelings, listen, reward. Check unhealthy relationships (smoking with friends, etc)

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On the other side, physical activity that includes habitual, moderate endurance exercise enhances immunity through enhanced lymphocytes, NK cell and total T-cell counts, increases lymphocyte proliferation and increases immunoglobulin levels.

Clinical Recommendations for Physical Activity

Children/Adolescents	Adults
- Cardiovascular and strength exercises (moderate-vigorous intensity) 60 min/daily ($\approx$ 300 min/wk) - Activity should be vigorous at LEAST 3 days/wk - Muscle and bone strengthening activities included in 60 minutes daily	- Cardiovascular and strength exercises (moderate intensity) 30 min/day ($\approx$ 150 min/wk) - OR 75 min/wk vigorous activity - Muscle-strengthening activity involving all major muscle groups at LEAST 2+ days/wk - Activity should be spread over at LEAST 3+ days/wk

Find the modality that works for your client

- Enjoyment, injury reduction, comfort, self-efficacy, social support

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